CM INTERSECTION MULTIPLICITIES EQUAL LENGTHS OF SCHEME-THEORETIC INTERSECTIONS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 090V: sources/stacks-chow/tags.txt, source lines 11423–11423. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00N0: sources/stacks-chow/tags.txt, source lines 842–842. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a nonsingular variety and $W, V \subset X$ closed subvarieties which intersect properly. Let Z be an irreducible component of $V \cap W$ with generic point ξ. Assume that $\mathcal{O}_{W,\xi}$ and $\mathcal{O}_{V,\xi}$ are Cohen-Macaulay. Then*

$$e(X, V \cdot W, Z) = \mathit{length}_{\mathcal{O}_{X,\xi}}(\mathcal{O}_{V \cap W,\xi})$$

where $V \cap W$ is the scheme theoretic intersection. In particular, if both V and W are Cohen-Macaulay, then $V \cdot W = [V \cap W]_{\dim(V)+\dim(W)-\dim(X)}$.

Proof. Set $A = \mathcal{O}_{X,\xi}$, $B = \mathcal{O}_{V,\xi}$, and $C = \mathcal{O}_{W,\xi}$. By Auslander-Buchsbaum (Algebra, Proposition 090V) we can find a finite free resolution $F_\bullet \to B$ of length

$$\operatorname{depth}(A) - \operatorname{depth}(B) = \dim(A) - \dim(B) = \dim(C)$$

First equality as A and B are Cohen-Macaulay and the second as V and W intersect properly. Then $F_\bullet \otimes_A C$ is a complex of finite free modules representing $B \otimes^{\mathbf{L}}_A C$ hence has cohomology modules with support in $\{\mathfrak{m}_A\}$. By the Acyclicity lemma (Algebra, Lemma 00N0) which applies as C is Cohen-Macaulay we conclude that $F_\bullet \otimes_A C$ has nonzero cohomology only in degree 0. This finishes the proof. □

Date: stacks-project:intersection.tex:local-snapshot-20261012; GNU FDL 1.2 or later.